

Proximal Balancing

Yonghan Jung

We study causal adjustment with observed pretreatment covariates X , a high-dimensional pretreatment proxy $W = (W_1, \dots, W_J)$, and an unobserved confounder. For one prespecified block j , the proposed cross-view construction learns a possibly stochastic, discrete or continuous representation $Z_j = \phi_j(V_j, \epsilon_j)$, with output dimension no larger than that of $V_j = (X, W_{-j})$, without using the held-out block W_j , and measures treatment imbalance through the conditional distribution of (X, W_j) . We prove an exact bias decomposition and show that exact cross-view balance identifies the population average treatment effect under joint latent exchangeability, held-out-view separation, overlap, and outcome-relevant completeness. When balance is approximate, a uniform outcome-relevant stability condition yields the sensitivity certificate $|\tau_{Z_j} - \tau| \leq \Gamma_j D_j(\phi_j)$. A finite-class, finite-strata learning analysis and an honest augmented inverse probability weighted estimator give an end-to-end finite-sample bound that separates representation error, hidden-confounding sensitivity, nuisance error, and evaluation-sample variation. The causal interpretation depends on the stated cross-view structural conditions; the existence of one generic correlated proxy is not sufficient by itself.

Contents

1. Introduction	3
1.1. Related Works	3
2. Problem Setting	4
3. Identification	4
3.1. Assumptions	5
3.2. Exact Cross-View Identification	6
3.3. Approximate Cross-View Bias Bound	8
3.4. Comparison with Existing Proximal Frameworks	9
4. Representation Learning and AIPW Estimation	9
4.1. Representation Learning	9
4.2. AIPW Estimation	13
4.3. End-to-End Finite-Sample Bound	14
5. Experiments	15
5.1. Questions, Comparators, and Reporting Rules	15

5.2. Honest Approximate SCMs	16
5.3. Exact-Assumption Tracks A and B	16
5.4. WSC Observational Benchmark	17
6. Discussion and Limitations	19
6.1. What the Method Delivers	19
6.2. Testable and Structural Components	19
6.3. Attainability and Realized Bias	19
6.4. Learning and Inference Gaps	19
6.5. Audit Views and Practical Scope	19
6.6. Scope of the Contribution	20
7. Conclusion	20
A. Simulation Details	22
B. Proofs	22
B.1. Proofs for Section 3	22
B.2. Proofs for Section 4	23

1. Introduction

1.1. Related Works

Balancing scores and covariate balancing. The propensity score is the canonical reduction of observed pretreatment covariates: under treatment ignorability given the measured covariates, conditioning on the treatment probability balances those covariates across treatment arms and identifies causal effects (Rosenbaum and Rubin, 1983). Entropy balancing and the covariate-balancing propensity score instead choose weights or treatment models that target balance directly (Hainmueller, 2012; Imai and Ratkovic, 2014). These methods motivate our optimization target, but their usual causal guarantee still requires the measured covariates to suffice for confounding control. PROBE instead allows a latent confounder and asks what a held-out proxy block can reveal about the bias remaining after reducing the other pretreatment measurements.

Causal representation learning. Representation-learning methods compress high-dimensional covariates while penalizing imbalance between treated and control representations. For example, counterfactual regression bounds counterfactual prediction error using factual prediction error and an integral probability metric between learned treatment-arm representations (Shalit et al., 2017). PROBE shares the use of a learned low-dimensional representation and a distributional balance penalty. Its distinguishing operation is blockwise exclusion: one proxy block is held out, the representation is learned from the remaining pretreatment information, and conditional treatment balance is evaluated with the held-out block.

Negative controls and proximal causal inference. Negative-control variables can reveal residual confounding because they should not be causally affected by treatment or should not causally affect the outcome, while remaining associated with hidden common causes (Lipsitch et al., 2010). Proximal causal inference turns this diagnostic idea into point identification by assigning distinct treatment-inducing and outcome-inducing proxy roles and imposing completeness and bridge conditions (Miao et al., 2018; Tchetgen Tchetgen et al., 2024). Semiparametric proximal theory then develops influence functions and efficient estimators for the resulting identifying functionals (Cui et al., 2024). PROBE takes a different operational route: the remaining proxy blocks form the input to an adjustment representation, while the held-out block measures residual treatment imbalance. An outcome-relevant completeness condition transfers exact observed balance to the potential-outcome means needed for the population average treatment effect.

Single-proxy identification. Single Proxy Control shows that one negative-control outcome can identify the effect of treatment on the treated under a counterfactual proxy condition and suitable bridge structure (Park et al., 2024). This differs from our population average treatment effect target, our latent-confounder proxy interpretation, and our use of an observable balance discrepancy. The comparison makes clear that “one proxy” does not by itself identify a common estimand. Identification depends on what the proxy measures, which conditional independence is imposed, and whether the analysis uses a bridge or an adjustment representation checked by a held-out proxy block.

Kernel discrepancies and sensitivity certification. Maximum mean discrepancy is an integral probability metric that equals zero only for identical distributions when its kernel is characteristic (Gretton et al., 2012). PROBE uses conditional treatment-arm discrepancies of the measured covariates and the held-out block as observable magnitudes, not kernel-test p -values. Multiplying this discrepancy by an outcome-relevant stability constant yields a bias certificate and a representation-

selection guarantee. The individual ingredients have close precedents in balancing, representation learning, kernel testing, and proximal inference. We therefore state the contribution through the precise assumptions, estimand, and guarantees below rather than making an unsupported priority claim.

2. Problem Setting

We observe independent copies

$$O_i = (X_i, W_i, A_i, Y_i), \quad i = 1, \dots, n, \quad (1)$$

where $X \in \mathcal{X}$ is a possibly high-dimensional vector of observed pretreatment covariates, $\mathcal{X}$ is its value space, $A \in \{0, 1\}$ is treatment, and Y is the observed outcome.

For each $a \in \{0, 1\}$, the *potential outcome* $Y(a)$ is the outcome that would be observed if treatment were set to a . Our target is the population average treatment effect

$$\tau \triangleq \mathbb{E}\{Y(1) - Y(0)\}. \quad (2)$$

Let U denote an unobserved pretreatment variable that may affect both treatment A and outcome Y . When U affects both A and Y , the observed-data distribution alone does not identify τ without additional structure. We therefore observe a proxy measurement W that contains information about U . The proxy is divided into prespecified blocks,

$$W = (W_1, \dots, W_J). \quad (3)$$

Each block may be scalar or vector-valued. A block can represent one repeated measurement, sensor channel, laboratory panel, or prespecified region of a larger measurement.

Fix one block index $j \in \{1, \dots, J\}$ before analyzing the data. We call W_j the *held-out proxy block* and let $\mathcal{W}_j$ denote its value space. We call the combination of all other blocks the *remaining proxy block* and write

$$W_{-j} \triangleq (W_1, \dots, W_{j-1}, W_{j+1}, \dots, W_J) \quad (4)$$

for the resulting vector of remaining proxy blocks. Let $\mathcal{V}_j$ denote the value space of the pair (X, W_{-j}) . Finally, define

$$V_j \triangleq (X, W_{-j}) \in \mathcal{V}_j. \quad (5)$$

The variable V_j is the combined vector of the observed covariates X and the remaining proxy blocks W_{-j} .

3. Identification

Let d_{V_j} denote the coordinate dimension of V_j . Fix a user-specified output dimension $d_{Z_j} \leq d_{V_j}$. Let $\epsilon_j \sim \text{Unif}(0, 1)$ be independent of $(U, V_j, W_j, A, Y(0), Y(1))$. For a measurable representation

function, define

$$\phi_j : \mathcal{V}_j \times [0, 1] \longrightarrow \mathbb{R}^{d_{Z_j}}, \quad Z_j \triangleq \phi_j(V_j, \epsilon_j). \quad (6)$$

The deterministic case is included when ϕ_j ignores its second input. Let Φ_j denote a prespecified class of such representation functions.

3.1. Assumptions

Assumption 1 (Consistency and integrability). *For each $a \in \{0, 1\}$,*

$$Y = Y(A), \quad \mathbb{E}|Y(a)| < \infty. \quad (7)$$

Consistency connects the observed outcome Y to the potential outcome corresponding to the treatment actually received. Integrability ensures that the causal means used below are finite.

Assumption 2 (Latent exchangeability and a common held-out channel). *For each $a \in \{0, 1\}$,*

$$Y(a) \perp\!\!\!\perp A \mid (U, V_j), \quad (8)$$

$$W_j \perp\!\!\!\perp A \mid (U, V_j). \quad (9)$$

Assumption 3 (Representation overlap).

$$0 < P(A = 1 \mid Z_j) < 1 \quad \text{almost surely.} \quad (10)$$

We then introduce the latent and proxy objects within each representation stratum.

Definition 1 (Latent and proxy objects within a representation stratum). *Let P_{Z_j} denote the distribution of Z_j . The conditional laws below refer to fixed versions that exist under the maintained probability model. For P_{Z_j} -almost every z , define*

$$Q_{a,j,z} \triangleq \mathcal{L}(V_j, U \mid A = a, Z_j = z), \quad a \in \{0, 1\}. \quad (11)$$

Define the held-out proxy channel by

$$K_j(\cdot \mid v, u) \triangleq \mathcal{L}(W_j \mid V_j = v, U = u). \quad (12)$$

For any probability law Q on (V_j, U) , let $\mathcal{K}_j Q$ denote the probability law on (X, W_j) generated in three steps. First draw $(V_j, U) = (X, W_{-j}, U) \sim Q$. Next draw $W_j \sim K_j(\cdot \mid V_j, U)$. Finally retain (X, W_j) and integrate out (W_{-j}, U) .

For each $a \in \{0, 1\}$, define

$$\mu_{a,j,z}(v, u) \triangleq \mathbb{E}\{Y(a) \mid V_j = v, U = u, Z_j = z\}. \quad (13)$$

The admissible law class is

$$\mathcal{Q}_{j,z} \triangleq \left\{ Q : Q \text{ is a probability law on } (V_j, U), Q \ll \mathcal{L}(V_j, U \mid Z_j = z), \right. \\ \left. \mathbb{E}_Q |\mu_{a,j,z}(V_j, U)| < \infty \text{ for } a \in \{0, 1\} \right\}. \quad (14)$$

For $Q \in \mathcal{Q}_{j,z}$, write

$$\mathbb{E}_Q \{\mu_{a,j,z}(V_j, U)\} \triangleq \int \mu_{a,j,z}(v, u) Q(dv, du). \quad (15)$$

The notation $Q \ll \mathcal{L}(V_j, U \mid Z_j = z)$ means that every event impossible under the stratum law of (V_j, U) is also impossible under Q . This domination makes the conditional-law versions well defined under Q . Under representation overlap and integrability, the actual arm laws $Q_{0,j,z}$ and $Q_{1,j,z}$ belong to $\mathcal{Q}_{j,z}$.

Assumption 4 (Exact outcome-relevant completeness). For P_{Z_j} -almost every z , every $Q, Q' \in \mathcal{Q}_{j,z}$, and each $a \in \{0, 1\}$,

$$\mathcal{K}_j Q = \mathcal{K}_j Q' \implies \mathbb{E}_Q \{\mu_{a,j,z}(V_j, U)\} = \mathbb{E}_{Q'} \{\mu_{a,j,z}(V_j, U)\}. \quad (16)$$

This condition is imposed on the entire class $\mathcal{Q}_{j,z}$. It need not imply $Q = Q'$ and therefore does not imply latent balance. On a common domain it is weaker than full mixture injectivity, but it is not uniformly comparable with standard proximal completeness because the operators differ.

3.2. Exact Cross-View Identification

Because Z_j is constructed from V_j and the independent variable ϵ_j , the next lemma shows that the two conditional independences in Assumption 2 remain valid after additionally conditioning on Z_j .

Lemma 1 (Conditioning safety). Under Assumption 2, for each $a \in \{0, 1\}$,

$$\mathbb{E}\{Y(a) \mid A, V_j, U, Z_j\} = \mathbb{E}\{Y(a) \mid V_j, U, Z_j\}, \quad (17)$$

and, for every measurable C ,

$$P(W_j \in C \mid A, V_j, U, Z_j) = K_j(C \mid V_j, U). \quad (18)$$

The lemma is used both to connect observed treatment-arm outcomes to the latent laws and to factorize the held-out proxy distribution.

Definition 2 (Adjustment functional). Let

$$m_{a,j}(z) \triangleq \mathbb{E}(Y \mid A = a, Z_j = z) \quad (19)$$

denote a version of the observed conditional outcome mean. Define

$$\mathbb{E}\{|m_{1,j}(Z_j)| + |m_{0,j}(Z_j)|\} < \infty. \quad (20)$$

Under this integrability condition, define

$$\tau_{Z_j} \triangleq \mathbb{E}\{m_{1,j}(Z_j) - m_{0,j}(Z_j)\}. \quad (21)$$

Lemma 2 (Cross-view factorization). *For each treatment arm and P_{Z_j} -almost every z ,*

$$\mathcal{L}(X, W_j \mid A = a, Z_j = z) = \mathcal{K}_j Q_{a,j,z}. \quad (22)$$

Definition 3 (Exact PROBE balance). *The fixed representation satisfies exact PROBE balance when*

$$(X, W_j) \perp\!\!\!\perp A \mid Z_j. \quad (23)$$

Exact balance and Lemma 2 imply, for P_{Z_j} -almost every z ,

$$\mathcal{K}_j Q_{1,j,z} = \mathcal{K}_j Q_{0,j,z}. \quad (24)$$

Theorem 1 (Exact identification under PROBE balance). *Suppose Assumptions 1–4 and equation (20) hold. If Z_j satisfies exact PROBE balance, then, for each $a \in \{0, 1\}$,*

$$\mathbb{E}\{Y(a) \mid A = 1, Z_j\} = \mathbb{E}\{Y(a) \mid A = 0, Z_j\} \quad \text{almost surely}, \quad (25)$$

and

$$\boxed{\tau_{Z_j} = \tau}. \quad (26)$$

Remark 1 (Why copying the representation input is not automatic). *For the non-compressive benchmark $Z_j = V_j$, balance of X is automatic, but held-out balance still requires $W_j \perp\!\!\!\perp A \mid V_j$. Thus copying does not automatically satisfy the observed endpoint.*

Remark 2 (What is, and is not, identified). *The theorem establishes conditional mean exchangeability, not full conditional independence or equality of the latent arm laws.*

Remark 3 (Attainability). *The theorem is conditional on exact balance and does not guarantee that a map attaining it exists in Φ_j .*

3.3. Approximate Cross-View Bias Bound

Let κ_j be a bounded, measurable, characteristic kernel on (X, W_j) whose reproducing kernel Hilbert space $\mathcal{H}_j$ is separable. For probability laws P, P' on (X, W_j) , write

$$\text{MMD}_{\kappa_j}(P, P') \triangleq \|m_j(P) - m_j(P')\|_{\mathcal{H}_j}, \quad (27)$$

where $m_j(P) = \int \kappa_j(r, \cdot) P(dr)$.

Definition 4 (Cross-view discrepancy). *For P_{Z_j} -almost every z , define*

$$\Delta_j(z; \phi_j) \triangleq \text{MMD}_{\kappa_j} \{ \mathcal{L}(X, W_j \mid A = 1, Z_j = z), \mathcal{L}(X, W_j \mid A = 0, Z_j = z) \}, \quad (28)$$

and define

$$D_j(\phi_j) \triangleq \mathbb{E}\{\Delta_j(Z_j; \phi_j)\}. \quad (29)$$

Proposition 1 (Zero discrepancy and exact balance). *Under representation overlap,*

$$D_j(\phi_j) = 0 \iff (X, W_j) \perp\!\!\!\perp A \mid Z_j. \quad (30)$$

The factorization lemma gives, almost everywhere,

$$\Delta_j(z; \phi_j) = \text{MMD}_{\kappa_j}(\mathcal{K}_j Q_{1,j,z}, \mathcal{K}_j Q_{0,j,z}). \quad (31)$$

Assumption 5 (Uniform outcome-relevant stability). *There is a finite Γ_j such that, for P_{Z_j} -almost every z , every $a \in \{0, 1\}$, and all $Q, Q' \in \mathcal{Q}_{j,z}$,*

$$|\mathbb{E}_Q\{\mu_{a,j,z}\} - \mathbb{E}_{Q'}\{\mu_{a,j,z}\}| \leq \Gamma_j \text{MMD}_{\kappa_j}(\mathcal{K}_j Q, \mathcal{K}_j Q'). \quad (32)$$

Uniform stability converts a difference between held-out proxy distributions into a common upper bound on the corresponding difference in potential-outcome means.

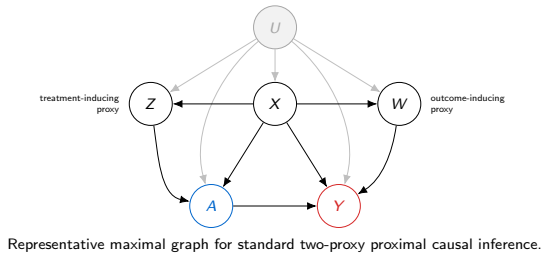
Theorem 2 (Approximate cross-view bias bound). *Under Assumptions 1–3, Assumption 5, and equation (20),*

$$|\tau_{Z_j} - \tau| \leq \Gamma_j D_j(\phi_j). \quad (33)$$

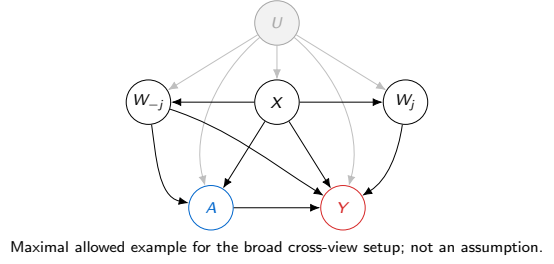
Remark 4 (Exact completeness and quantitative stability). *Stability implies exact outcome-relevant completeness, but the converse need not hold.*

Remark 5 (Sensitivity interpretation). *For a scientifically specified Γ_j ,*

$$\tau \in [\tau_{Z_j} - \Gamma_j D_j(\phi_j), \tau_{Z_j} + \Gamma_j D_j(\phi_j)]. \quad (34)$$



(a) Standard proximal



(b) PROBE-compatible

Figure 1: Illustrative graphs for standard proximal causal inference and PROBE. The graphs do not by themselves imply the conditional-independence, completeness, or stability assumptions.

Remark 6 (Uniformity across representations). *Learning by discrepancy requires a common budget*

$$\sup_{\phi \in \Phi_j} \Gamma_j(\phi) \leq \bar{\Gamma}_j < \infty. \quad (35)$$

3.4. Comparison with Existing Proximal Frameworks

Standard proximal causal inference assigns treatment-inducing and outcome-inducing proxy roles and uses bridge equations and completeness (Miao et al., 2018; Tchetgen Tchetgen et al., 2024). PROBE instead holds out W_j , constructs Z_j from V_j , and links observed balance to causal conclusions through outcome-relevant completeness or stability.

4. Representation Learning and AIPW Estimation

The identification results concern one fixed representation. We now select a representation from data and estimate the adjustment functional that it induces. The analysis uses one prespecified held-out block j throughout. The AIPW construction below applies to discrete or continuous Z_j . Explicit concentration constants are subsequently given for a finite-strata specialization.

4.1. Representation Learning

Partition the observed sample into three mutually disjoint subsets,

$$\mathcal{I}_D, \quad \mathcal{I}_N, \quad \mathcal{I}_E, \quad (36)$$

with respective sizes n_D , n_N , and n_E . The discrepancy-learning sample $\mathcal{I}_D$ selects a representation. The nuisance-learning sample $\mathcal{I}_N$ estimates the outcome regressions and propensity used by AIPW. The evaluation sample $\mathcal{I}_E$ calculates the final estimate. Conditional on the preceding samples, the objects evaluated on the next sample are fixed.

Augment each fold with fresh independent randomization variables $\epsilon_{j,i}^D$, $\epsilon_{j,i}^N$, and $\epsilon_{j,i}^E$, each distributed as $\text{Unif}(0, 1)$ and independent of the observed data and of all other randomization variables. For a candidate ϕ , its representation on a unit in fold $s \in \{D, N, E\}$ is $\phi(V_{j,i}, \epsilon_{j,i}^s)$. Thus the empirical criterion below is computed from the augmented discrepancy-learning sample $\{(O_i, \epsilon_{j,i}^D) : i \in \mathcal{I}_D\}$.

For a general candidate class $\mathcal{F}_j \subseteq \Phi_j$, let $\hat{D}_j(\phi)$ be an empirical criterion estimating $D_j(\phi)$. Suppose that, with probability at least $1 - \delta_D$,

$$\sup_{\phi \in \mathcal{F}_j} |\hat{D}_j(\phi) - D_j(\phi)| \leq r_{j,n_D}. \quad (37)$$

If $\hat{\phi}_j$ is an $\varepsilon_{\text{opt},j}$ -approximate minimizer of $\hat{D}_j$, the standard empirical-risk argument gives

$$D_j(\hat{\phi}_j) \leq \inf_{\phi \in \mathcal{F}_j} D_j(\phi) + 2r_{j,n_D} + \varepsilon_{\text{opt},j}. \quad (38)$$

This oracle inequality is agnostic to whether the learned representation is discrete or continuous. Its operational content depends on constructing an empirical conditional-dependence criterion for which equation (37) can be proved. For a fresh $\epsilon_j \sim \text{Unif}(0, 1)$ independent of the causal variables, define

$$\hat{Z}_j \triangleq \hat{\phi}_j(V_j, \epsilon_j), \quad \tau_{\hat{Z}_j} \triangleq \mathbb{E}[\mathbb{E}(Y \mid A = 1, \hat{Z}_j) - \mathbb{E}(Y \mid A = 0, \hat{Z}_j)]. \quad (39)$$

This definition is used when the two conditional means are absolutely integrable under the law of $\hat{Z}_j$. If every candidate obeys the assumptions of Theorem 2 with common stability budget $\bar{\Gamma}_j$, then the same event also gives

$$|\tau_{\hat{Z}_j} - \tau| \leq \bar{\Gamma}_j \left\{ \inf_{\phi \in \mathcal{F}_j} D_j(\phi) + 2r_{j,n_D} + \varepsilon_{\text{opt},j} \right\}. \quad (40)$$

Finite-strata specialization. We now give one explicit realization of the preceding argument. Restrict to deterministic maps, identifying each such map with $v \mapsto \phi(v, 0)$ and writing it as $\phi(v)$. Assume every candidate takes values in a common finite set $\mathcal{S}_j \subset \mathbb{R}^{d_{Z_j}}$, and define $M_j \triangleq |\mathcal{S}_j|$.

Fix a finite candidate library

$$\mathcal{F}_j \subseteq \Phi_j, \quad N_j \triangleq |\mathcal{F}_j| < \infty. \quad (41)$$

For $\phi \in \mathcal{F}_j$, define

$$Z_j^\phi \triangleq \phi(X, W_{-j}) \quad (42)$$

and let $\mathcal{R}_j(\phi)$ be its set of positive-mass strata. For $z \in \mathcal{R}_j(\phi)$, write $p_{j,z}(\phi) \triangleq P(Z_j^\phi = z)$. Assume

there are constants $p_{\min} > 0$ and $\eta \in (0, 1/2]$ such that every candidate and every $z \in \mathcal{R}_j(\phi)$ satisfy

$$P(Z_j^\phi = z) \geq p_{\min}, \quad (43)$$

$$\eta \leq P(A = 1 \mid Z_j^\phi = z) \leq 1 - \eta. \quad (44)$$

Assume also that the bounded kernel satisfies

$$\sup_v \sqrt{\kappa_j(v, v)} \leq \rho_j < \infty. \quad (45)$$

The two primitive cross-view clauses of Assumption 2 do not depend on the selected candidate. Representation overlap and outcome-relevant stability must hold for every candidate-specific representation. Their stability budgets are required to obey the common bound in equation (35).

Definition 5 (Empirical cross-view discrepancy). *For observation $i \in \mathcal{I}_D$ and candidate $\phi \in \mathcal{F}_j$, write*

$$Z_{j,i}^\phi \triangleq \phi(X_i, W_{-j,i}). \quad (46)$$

For treatment arm a and stratum z , define

$$N_{a,z}^\phi \triangleq \sum_{i \in \mathcal{I}_D} 1\{A_i = a, Z_{j,i}^\phi = z\}, \quad (47)$$

$$N_z^\phi \triangleq N_{0,z}^\phi + N_{1,z}^\phi, \quad (48)$$

$$\hat{p}_{j,z}(\phi) \triangleq \frac{N_z^\phi}{n_D}. \quad (49)$$

When $N_{a,z}^\phi > 0$, define the empirical kernel mean

$$\hat{m}_{a,j,z}(\phi) \triangleq \frac{1}{N_{a,z}^\phi} \sum_{i \in \mathcal{I}_D} 1\{A_i = a, Z_{j,i}^\phi = z\} \kappa_j\{(X_i, W_{j,i}), \cdot\}. \quad (50)$$

When both treatment-arm cells are nonempty, define

$$\hat{\Delta}_{j,z}(\phi) \triangleq \|\hat{m}_{1,j,z}(\phi) - \hat{m}_{0,j,z}(\phi)\|_{\mathcal{H}_j} \quad (51)$$

and

$$\hat{D}_j(\phi) \triangleq \sum_{z: N_z^\phi > 0} \hat{p}_{j,z}(\phi) \hat{\Delta}_{j,z}(\phi). \quad (52)$$

If an occupied empirical stratum contains only one treatment arm, set $\hat{D}_j(\phi) = +\infty$.

Proposition 2 (Uniform deviation for the finite-strata specialization). *Let $\delta_D \in (0, 1)$ and suppose*

$$n_D \geq \frac{8}{p_{\min} \eta} \log \left(\frac{8M_j N_j}{\delta_D} \right). \quad (53)$$

Define

$$e_{j,n_D}(\delta_D) \triangleq \rho_j \left[2 + \sqrt{2 \log \left(\frac{4M_j N_j}{\delta_D} \right)} \right] \sqrt{\frac{2}{n_D p_{\min} \eta}} \quad (54)$$

and

$$\begin{aligned} r_{j,n_D}(\delta_D) &\triangleq 2e_{j,n_D}(\delta_D) \\ &\quad + 2\rho_j \sqrt{\frac{2 \left[M_j \log 2 + \log \left(\frac{4N_j}{\delta_D} \right) \right]}{n_D}}. \end{aligned} \quad (55)$$

Under equations (43) through (45), with probability at least $1 - \delta_D$,

$$\boxed{\sup_{\phi \in \mathcal{F}_j} |\hat{D}_j(\phi) - D_j(\phi)| \leq r_{j,n_D}(\delta_D).} \quad (56)$$

Definition 6 (Approximate empirical minimizer). *Let $\varepsilon_{\text{opt},j} \geq 0$. The representation learner returns $\hat{\phi}_j \in \mathcal{F}_j$ satisfying*

$$\hat{D}_j(\hat{\phi}_j) \leq \inf_{\phi \in \mathcal{F}_j} \hat{D}_j(\phi) + \varepsilon_{\text{opt},j}. \quad (57)$$

In this deterministic specialization, the learned representation is

$$\hat{Z}_j \triangleq \hat{\phi}_j(V_j). \quad (58)$$

The tolerance records optimization error and is zero for an exact empirical minimizer.

Theorem 3 (Finite-strata representation-learning oracle inequality). *Suppose the conditions of Proposition 2 hold. Suppose Assumptions 1 through 2 hold for block j , and suppose representation overlap and uniform outcome-relevant stability hold for every candidate with the common budget $\bar{\Gamma}_j$. Then, with probability at least $1 - \delta_D$,*

$$\boxed{D_j(\hat{\phi}_j) \leq \inf_{\phi \in \mathcal{F}_j} D_j(\phi) + 2r_{j,n_D}(\delta_D) + \varepsilon_{\text{opt},j}.} \quad (59)$$

On the same event,

$$\boxed{|\tau_{\hat{Z}_j} - \tau| \leq \bar{\Gamma}_j \left[\inf_{\phi \in \mathcal{F}_j} D_j(\phi) + 2r_{j,n_D}(\delta_D) + \varepsilon_{\text{opt},j} \right].} \quad (60)$$

Remark 7 (Practical learner outside the finite-library theorem). *The theorem covers hard, finite-valued representations in a finite prespecified library. A neural learner using random features, soft assignments, and Adam requires additional bounds for neural-class complexity, feature approximation, soft-to-hard transfer, and optimization error. These guarantees are*

not supplied by Theorem 3.

4.2. AIPW Estimation

Conditional on $\mathcal{I}_D$, the map $\hat{\phi}_j$ is fixed. A fresh independent $\epsilon_j \sim \text{Unif}(0, 1)$ produces $\hat{Z}_j = \hat{\phi}_j(V_j, \epsilon_j)$. Define the conditional outcome means and representation propensity

$$m_{a,j}(z) \triangleq \mathbb{E}(Y \mid A = a, \hat{Z}_j = z), \quad (61)$$

$$e_j(z) \triangleq P(A = 1 \mid \hat{Z}_j = z). \quad (62)$$

For the observed adjustment functional $\tau_{\hat{Z}_j}$, the following is the AIPW score:

$$\begin{aligned} \psi_j(O, \epsilon_j) &\triangleq m_{1,j}(\hat{Z}_j) - m_{0,j}(\hat{Z}_j) \\ &+ \frac{A\{Y - m_{1,j}(\hat{Z}_j)\}}{e_j(\hat{Z}_j)} - \frac{(1 - A)\{Y - m_{0,j}(\hat{Z}_j)\}}{1 - e_j(\hat{Z}_j)} - \tau_{\hat{Z}_j}. \end{aligned} \quad (63)$$

When this score is square-integrable, it is the efficient influence function in the nonparametric model for $(\hat{Z}_j, A, Y)$. Bounded outcomes together with uniform representation overlap provide one sufficient condition for square integrability. Use the augmented nuisance-learning sample, with $\hat{Z}_{j,i} = \hat{\phi}_j(V_{j,i}, \epsilon_{j,i}^N)$, to estimate the nuisance functions. Denote the estimates by $\hat{m}_{0,j}$, $\hat{m}_{1,j}$, and $\hat{e}_j$.

Definition 7 (Honest AIPW estimator). For $i \in \mathcal{I}_E$, define $\hat{Z}_{j,i} = \hat{\phi}_j(V_{j,i}, \epsilon_{j,i}^E)$ and

$$\begin{aligned} S_j(O_i, \epsilon_{j,i}^E) &\triangleq \hat{m}_{1,j}(\hat{Z}_{j,i}) - \hat{m}_{0,j}(\hat{Z}_{j,i}) \\ &+ \frac{A_i\{Y_i - \hat{m}_{1,j}(\hat{Z}_{j,i})\}}{\hat{e}_j(\hat{Z}_{j,i})} \\ &- \frac{(1 - A_i)\{Y_i - \hat{m}_{0,j}(\hat{Z}_{j,i})\}}{1 - \hat{e}_j(\hat{Z}_{j,i})}. \end{aligned} \quad (64)$$

The honest AIPW estimator is

$$\hat{\tau}_j^{\text{AIPW}} \triangleq \frac{1}{n_E} \sum_{i \in \mathcal{I}_E} S_j(O_i, \epsilon_{j,i}^E). \quad (65)$$

Proposition 3 (Exact signed nuisance remainder). Conditional on $\mathcal{I}_D$ and $\mathcal{I}_N$,

$$\begin{aligned} &\mathbb{E}\{S_j(O, \epsilon_j)\} - \tau_{\hat{Z}_j} \\ &= \mathbb{E} \left[\left\{ \hat{e}_j(\hat{Z}_j) - e_j(\hat{Z}_j) \right\} \left\{ \frac{\hat{m}_{1,j}(\hat{Z}_j) - m_{1,j}(\hat{Z}_j)}{\hat{e}_j(\hat{Z}_j)} \right. \right. \\ &\quad \left. \left. + \frac{\hat{m}_{0,j}(\hat{Z}_j) - m_{0,j}(\hat{Z}_j)}{1 - \hat{e}_j(\hat{Z}_j)} \right\} \right]. \end{aligned} \quad (66)$$

The remainder is zero if the propensity estimator is correct. It is also zero if both conditional outcome estimators are correct.

Theorem 4 (Conditional finite-sample AIPW error). *Suppose $|Y| \leq B$ almost surely. Clip the nuisance estimates so that*

$$-B \leq \widehat{m}_{a,j}(z) \leq B, \quad \eta \leq \widehat{e}_j(z) \leq 1 - \eta. \quad (67)$$

For a function g of $\widehat{Z}_j$, define

$$\|g\|_{2,\widehat{Z}_j} \triangleq \left[\mathbb{E}\{g(\widehat{Z}_j)^2 \mid \mathcal{I}_D, \mathcal{I}_N\} \right]^{1/2}. \quad (68)$$

After the two learning samples are observed, define

$$q_{e,j} \triangleq \|\widehat{e}_j - e_j\|_{2,\widehat{Z}_j}, \quad (69)$$

$$q_{a,j} \triangleq \|\widehat{m}_{a,j} - m_{a,j}\|_{2,\widehat{Z}_j}, \quad a \in \{0, 1\}. \quad (70)$$

For $\delta_E \in (0, 1)$, define

$$\begin{aligned} \varepsilon_{\text{AIPW},j} &\triangleq 2B \left(1 + \frac{1}{\eta}\right) \sqrt{\frac{2 \log(2/\delta_E)}{n_E}} \\ &\quad + \frac{q_{e,j}(q_{0,j} + q_{1,j})}{\eta}. \end{aligned} \quad (71)$$

Conditional on $\mathcal{I}_D$ and $\mathcal{I}_N$, with probability at least $1 - \delta_E$ over $\mathcal{I}_E$,

$$|\widehat{\tau}_j^{\text{AIPW}} - \tau_{\widehat{Z}_j}| \leq \varepsilon_{\text{AIPW},j}. \quad (72)$$

4.3. End-to-End Finite-Sample Bound

Theorem 5 (Finite-strata end-to-end sensitivity bound). *Suppose the conditions of Proposition 2, Theorem 3, and Theorem 4 hold. Then, jointly over the three independent samples, with probability at least $1 - \delta_D - \delta_E$,*

$$|\widehat{\tau}_j^{\text{AIPW}} - \tau| \leq \varepsilon_{\text{AIPW},j} + \bar{\Gamma}_j \left[\widehat{D}_j(\widehat{\phi}_j) + r_{j,n_D}(\delta_D) \right]. \quad (73)$$

It also satisfies the oracle form

$$\begin{aligned} |\widehat{\tau}_j^{\text{AIPW}} - \tau| &\leq \varepsilon_{\text{AIPW},j} \\ &\quad + \bar{\Gamma}_j \left[\inf_{\phi \in \mathcal{F}_j} D_j(\phi) + 2r_{j,n_D}(\delta_D) + \varepsilon_{\text{opt},j} \right]. \end{aligned} \quad (74)$$

Corollary 1 (Consistency). *Consider a sequence of candidate libraries, sample splits, nuisance estimators, optimization tolerances, and stability budgets satisfying Theorem 5. If*

$$\delta_{D,n} + \delta_{E,n} \longrightarrow 0, \quad (75)$$

$$\frac{\log(2/\delta_{E,n})}{n_E} \longrightarrow 0, \quad (76)$$

$$q_{e,j,n}(q_{0,j,n} + q_{1,j,n}) \xrightarrow{p} 0, \quad (77)$$

and

$$\bar{\Gamma}_{j,n} \left[\inf_{\phi \in \mathcal{F}_{j,n}} D_j(\phi) + 2r_{j,n_D}(\delta_{D,n}) + \varepsilon_{\text{opt},j,n} \right] \xrightarrow{p} 0, \quad (78)$$

then

$$\hat{\tau}_{j,n}^{\text{AIPW}} \xrightarrow{p} \tau. \quad (79)$$

Remark 8 (Interpretation and observability). *Theorem 5 separates the final error into an AIPW estimation term and a hidden-confounding sensitivity term. The stability budget $\bar{\Gamma}_j$ and the nuisance errors $q_{e,j}$, $q_{0,j}$, and $q_{1,j}$ are not generally observed. The result becomes an operational numerical certificate after these quantities are bounded by additional theory, external information, or a declared sensitivity analysis.*

Remark 9 (Scope). *The theorem uses one prespecified held-out block, hard finite strata, a finite candidate library, and an honest three-way split. It does not establish a neural-class rate, a soft-to-hard approximation bound, an Adam optimization guarantee, or a cross-fitting theorem. Experimental aggregation across several held-out orientations is also outside this fixed- j result.*

5. Experiments

5.1. Questions, Comparators, and Reporting Rules

The experiments answer three separate questions. First, does proxy-supervised representation adjustment reduce ATE error relative to adjustment using X alone? Second, do the exact-assumption designs recover the known balancing structure? Third, does the method show a useful descriptive signal in an observational benchmark with a noisy randomized reference? We report mean absolute error, root mean squared error, paired wins, held-out discrepancy, overlap diagnostics, and Monte Carlo uncertainty. Oracle procedures that observe U or the true balancing score are simulation references, not implementable competitors. The theory in Sections 3 and 4 applies to one prespecified held-out block at a time. Experimental aggregation across several orientations is reported as an empirical extension and is not presented as a separate identification theorem.

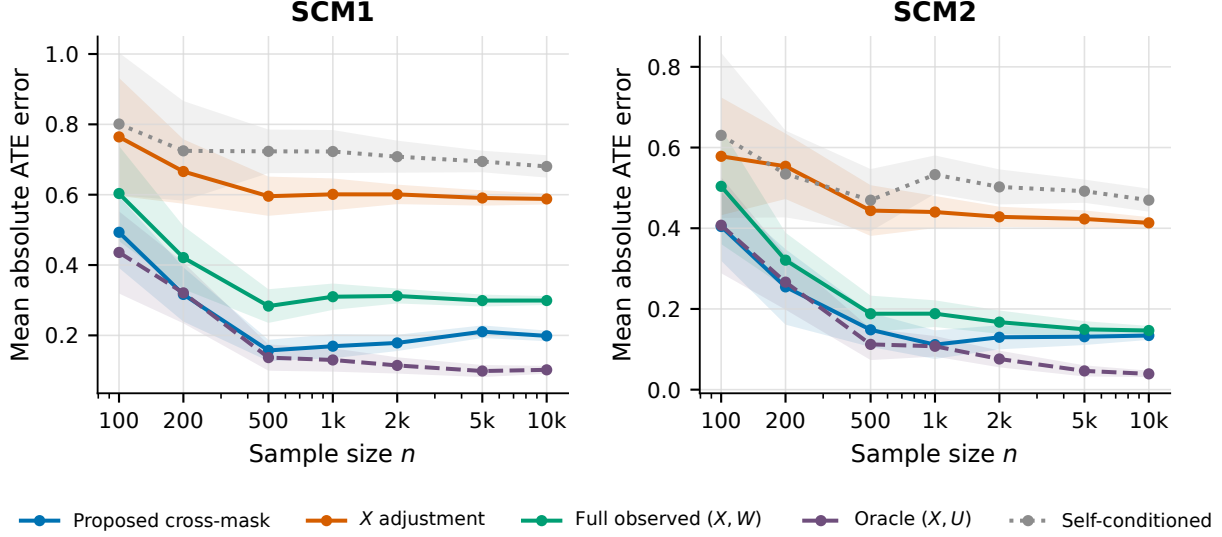


Figure 2: Honest approximate SCMs. Curves show mean absolute ATE error across twenty independently generated data sets; bands are approximate 95% Monte Carlo intervals for the mean. Prefixes are nested within a data set, so uncertainty is computed separately at each n and is not pooled across sample sizes. The proposed cross-mask learner uses one proxy coordinate for held-out auditing and averages across masks.

5.2. Honest Approximate SCMs

The primary approximate study fixes the causal order $U \rightarrow (X, W)$, $(X, U) \rightarrow A$, and $(A, X, U) \rightarrow Y$. Both SCMs use $\dim(X) = 20$ and $\dim(W) = 10$, with twenty independently generated data sets and nested prefixes $n \in \{100, 200, 500, 1000, 2000, 5000, 10000\}$. SCM1 uses linear proxy measurements and a nonlinear outcome component; SCM2 changes the structural equations to nonlinear transformations while preserving the same causal order and an informative monotone proxy channel. The proposed cross-mask method learns a representation from X and all but one proxy coordinate, audits balance with the held-out coordinate, and averages over masks.

At $n = 1000$, the mean absolute errors of proposed/ X -adjustment/full- (X, W) /oracle- (X, U) were 0.169/0.601/0.310/0.130 in SCM1 and 0.112/0.440/0.188/0.107 in SCM2. At $n = 10000$, the corresponding proposed errors were 0.199 and 0.134, versus 0.588 and 0.413 for X adjustment. These are finite-sample simulation results, not a proof of the audit-view identification theorem. The self-conditioned heuristic remained substantially biased, which is consistent with the self-conditioning warning in Section 3.

5.3. Exact-Assumption Tracks A and B

Track A uses a finite balancing score, $\dim(X) = 20$, nine representation-side proxy coordinates, and one independent audit coordinate. The DGP enforces strict population overlap, exact decoding of the discrete confounding state from the representation-side view, and an injective Gaussian audit channel. At $n = 10000$ over twenty replications, mean absolute error was 0.175 for the learned audit representation, 0.433 for X adjustment, 0.186 for direct full-observed adjustment, and 0.033 for the

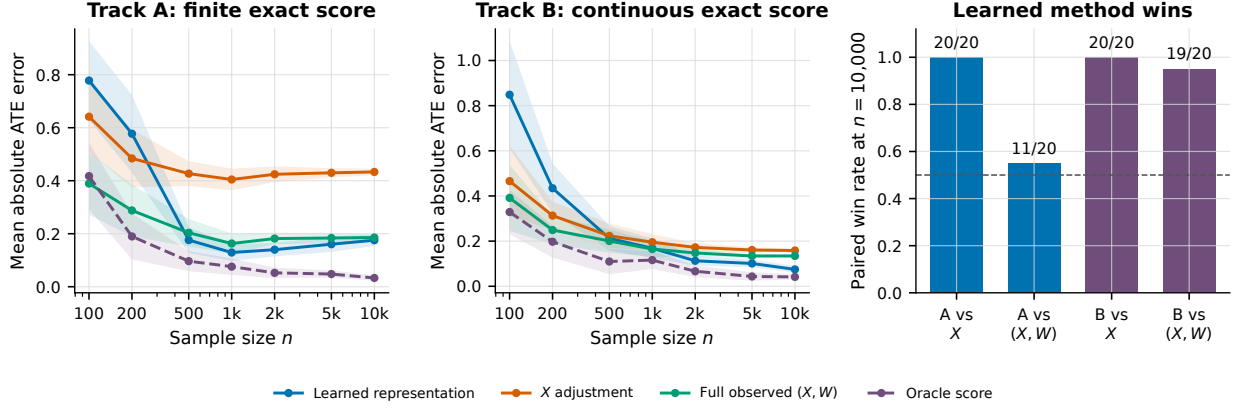


Figure 3: Exact-assumption experiments. The first two panels report mean absolute ATE error with approximate 95% Monte Carlo intervals over twenty replications. The third panel reports paired win rates at $n = 10,000$, where a win means that the learned representation has smaller absolute error than the indicated comparator. Track A satisfies the primary comparison with X but not the prespecified secondary superiority criterion against full observed adjustment. Track B records rank-recovery Spearman correlation 0.915 and held-out finite-RFF conditional-dependence surrogate 0.00691.

oracle balancing score. The primary comparison with X passed, while the prespecified secondary superiority criterion against full-observed adjustment failed.

Track B uses a continuous exact balancing score and a linear representation class matched to that score. At $n = 10000$, mean absolute error was 0.0748 for the learned representation, 0.1583 for fitted X adjustment, and 0.1342 for fitted full-observed adjustment. The learned method beat X in 20/20 and full-observed adjustment in 19/20 paired replications. Its mean rank-recovery Spearman correlation was 0.915. The held-out conditional-dependence diagnostic used finite random features and is a surrogate, not an exact population KCI test; the design also covers one prespecified scalar- U SCM family.

5.4. WSC Observational Benchmark

The WSC study forms an observational subset of 1,105 participants whose randomized mathematics assignment agrees with pretreatment preference, and targets the population ATE in the original 2,200 randomized participants. The full randomized difference in means is 0.760 with standard error 0.151 and serves only as a noisy benchmark. Across ten algorithmic split repetitions on this one fixed data set, the unconstrained cross-mask estimate averaged 1.302 with split standard deviation 0.065 and absolute benchmark gap 0.541, compared with gaps 2.690 for the naive observational contrast, 1.704 for direct X adjustment, and 0.672 for direct (X, W) adjustment. The unconstrained cross-mask method is the proposed WSC method shown in Figure 4. Its practical-overlap gate passed 0/10 repetitions. The benchmark therefore supplies descriptive evidence only, not a validated causal-performance claim, causal ground truth, or a test of proxy validity.

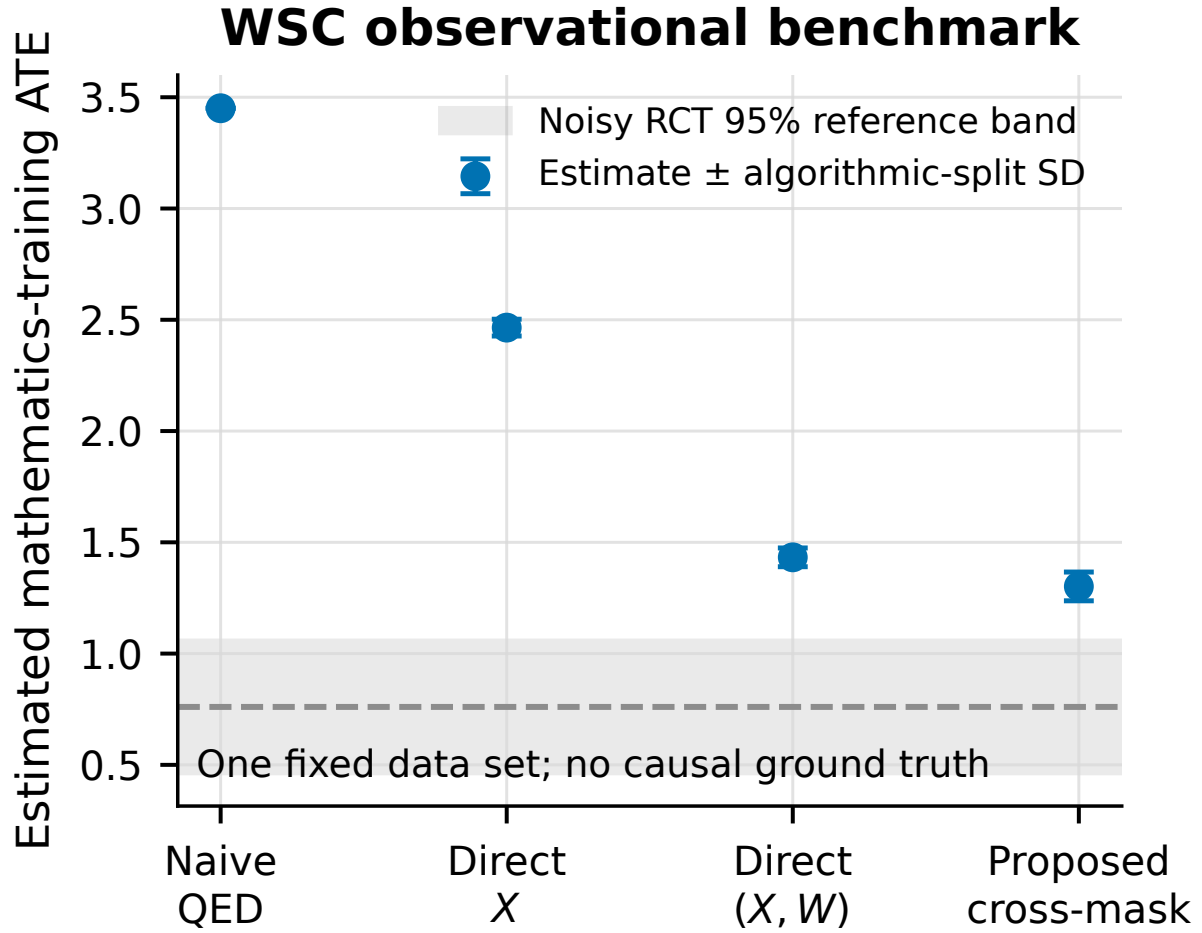


Figure 4: WSC observational benchmark. Estimates are shown relative to the noisy randomized reference. Error bars for fitted methods are standard deviations across ten algorithmic splits of one fixed data set, not Monte Carlo uncertainty. The benchmark has no causal ground truth and reports only the unconstrained cross-mask learner as the proposed method.

6. Discussion and Limitations

6.1. What the Method Delivers

For one prespecified block j , the canonical method uses W_{-j} to construct an outcome-free representation and uses the held-out block W_j to measure the residual imbalance of (X, W_j) across treatment arms within that representation. Under uniform outcome-relevant stability, the product $\Gamma_j D_j(\phi_j)$ is a sensitivity certificate for hidden-confounding bias. This is stronger than reporting a conditional-independence test alone because it targets a causal error scale, but weaker than point identification when $D_j(\phi_j) > 0$ because Γ_j is not generally identified from the observed distribution.

6.2. Testable and Structural Components

Held-out joint balance, empirical overlap, and minimum stratum mass are observable properties. Joint latent exchangeability, separation of the held-out proxy block, outcome-relevant completeness, and outcome-relevant stability are structural assumptions about the full-data law. Exact balance is therefore an observable population condition that can only be assessed up to sampling error, while its causal interpretation still requires structural assumptions. A large held-out discrepancy can falsify exact balance; a small test statistic cannot prove exact balance, completeness, or stability.

6.3. Attainability and Realized Bias

The exact identification theorem is conditional on the existence of a representation $Z_j = \phi_j(V_j, \epsilon_j)$ that attains exact cross-view balance. The theorem does not guarantee such attainability in a restricted representation class. In the approximate regime, reducing $D_j(\phi_j)$ reduces the certified worst-case bias under a common stability budget, but it need not reduce the realized bias when candidate representations have different stability constants or when the structural stability condition fails.

6.4. Learning and Inference Gaps

The current explicit finite-sample theorem covers finite candidate libraries with hard finite strata and fixed-fold evaluation. The neural experiments use soft assignments and covariance-weighted surrogate objectives that are not algebraically identical to the canonical MMD aggregation. A neural-class generalization bound, a soft-to-hard guarantee, and valid joint inference after adaptive representation selection remain open. These gaps prevent the manuscript from describing every neural result as a direct verification of the theory, even when its empirical error is lower.

6.5. Audit Views and Practical Scope

Cross-view learning uses one prespecified proxy block for held-out evaluation and the remaining high-dimensional measurement for representation learning. This design blocks literal self-copying of the evaluated block. It does not by itself establish separation among proxy views, so point identification still needs the structural assumptions in Section 3. The experiments repeat the procedure

across several held-out orientations and aggregate their estimates as an empirical extension; the fixed- j theory applies separately to each orientation. The WSC experiment further shows that an encouraging benchmark gap can coexist with a failed practical-overlap rule. Real-data evidence must therefore report overlap, held-out discrepancies, and benchmark uncertainty without treating any one diagnostic as ground truth.

6.6. Scope of the Contribution

The defensible contribution is the combination of a cross-view proxy-supervised representation, an observable conditional discrepancy, an exact identification theorem, a stability-indexed approximate guarantee, and an honest finite-sample estimation analysis. The paper does not claim that one generic correlated proxy point-identifies the ATE, that every smaller discrepancy has smaller realized bias without a common stability budget, or that the current neural optimizer reaches the population optimum. A systematic literature audit and broader external validation are required before any firstness or universality claim.

7. Conclusion

Single proxy balancing asks what a structured high-dimensional pretreatment proxy can reveal about hidden-confounding bias after learning a low-dimensional adjustment representation. For one prespecified held-out block j , the cross-view construction learns $Z_j = \phi_j(V_j, \epsilon_j)$ and evaluates treatment balance through (X, W_j) . Exact cross-view balance identifies the ATE under the stated structural assumptions, while approximate balance yields the sensitivity certificate $|\tau_{Z_j} - \tau| \leq \Gamma_j D_j(\phi_j)$. Empirical discrepancy minimization and honest AIPW estimation give an end-to-end bound that separates learning, hidden-confounding, nuisance, and evaluation errors. The current simulations support the method in exact and approximate designs while exposing limitations from unattainable balance, weak proxy channels, surrogate mismatch, and practical-overlap loss. Neural-class generalization, soft-to-hard transfer, stability calibration, and external validation remain necessary for a submission-ready paper.

References

- Cui, Y., Pu, H., Shi, X., Miao, W., and Tchetgen Tchetgen, E. (2024). Semiparametric proximal causal inference. *Journal of the American Statistical Association*, 119(546):1348–1359.
- Gretton, A., Borgwardt, K. M., Rasch, M. J., Schölkopf, B., and Smola, A. (2012). A kernel two-sample test. *Journal of Machine Learning Research*, 13:723–773.
- Hainmueller, J. (2012). Entropy balancing for causal effects: A multivariate reweighting method to produce balanced samples in observational studies. *Political Analysis*, 20(1):25–46.
- Imai, K. and Ratkovic, M. (2014). Covariate balancing propensity score. *Journal of the Royal Statistical Society: Series B*, 76(1):243–263.
- Lipsitch, M., Tchetgen Tchetgen, E., and Cohen, T. (2010). Negative controls: A tool for detecting confounding and bias in observational studies. *Epidemiology*, 21(3):383–388.
- Miao, W., Geng, Z., and Tchetgen Tchetgen, E. J. (2018). Identifying causal effects with proxy variables of an unmeasured confounder. *Biometrika*, 105(4):987–993.
- Park, C., Richardson, D. B., and Tchetgen Tchetgen, E. J. (2024). Single proxy control. *Biometrics*, 80(2):ujae027.
- Rosenbaum, P. R. and Rubin, D. B. (1983). The central role of the propensity score in observational studies for causal effects. *Biometrika*, 70(1):41–55.
- Shalit, U., Johansson, F. D., and Sontag, D. (2017). Estimating individual treatment effect: Generalization bounds and algorithms. In *Proceedings of the 34th International Conference on Machine Learning*, volume 70 of *Proceedings of Machine Learning Research*, pages 3076–3085.
- Tchetgen Tchetgen, E. J., Ying, A., Cui, Y., Shi, X., and Miao, W. (2024). An introduction to proximal causal inference. *Statistical Science*, 39(3):375–390.

A. Simulation Details

The existing project-local experiments will be documented by recording each data-generating process, parameter value, sample size, replication count, and random seed policy. Every manuscript figure and table will be linked to the script and machine-readable result from which it was produced.

The simulation audit will separate observed quantities from oracle quantities. In particular, any result that uses U or a direction constructed from U will be labeled as an oracle diagnostic and will not be presented as the output of a learnable procedure.

B. Proofs

B.1. Proofs for Section 3

Conditioning safety.

Proof of Lemma 1. Conditional on V_j , Z_j depends only on ϵ_j . The displayed independence of ϵ_j makes adding Z_j safe in each of the two separate clauses of Assumption 2. $\square$

Cross-view factorization.

Proof of Lemma 2. Integrate the common conditional law in equation (18) over $Q_{a,j,z}$. By the generative definition of $\mathcal{K}_j$ in Definition 1, the resulting law of (X, W_j) is $\mathcal{K}_j Q_{a,j,z}$. $\square$

Exact identification.

Proof of Theorem 1. Exact PROBE balance and Lemma 2 imply

$$\mathcal{K}_j Q_{1,j,z} = \mathcal{K}_j Q_{0,j,z} \tag{80}$$

for P_{Z_j} -almost every z . Assumption 4 therefore gives, for each $a \in \{0, 1\}$,

$$\mathbb{E}_{Q_{1,j,z}} \{\mu_{a,j,z}(V_j, U)\} = \mathbb{E}_{Q_{0,j,z}} \{\mu_{a,j,z}(V_j, U)\}. \tag{81}$$

By Lemma 1 and iterated expectation, for each $b \in \{0, 1\}$,

$$\mathbb{E}\{Y(a) \mid A = b, Z_j = z\} = \mathbb{E}_{Q_{b,j,z}} \{\mu_{a,j,z}(V_j, U)\}. \tag{82}$$

Equation (81) proves equation (25). Consistency then gives

$$\begin{aligned} m_{a,j}(z) &= \mathbb{E}\{Y(a) \mid A = a, Z_j = z\} \\ &= \mathbb{E}\{Y(a) \mid Z_j = z\}. \end{aligned} \tag{83}$$

Integrating the difference between the two treatment levels with respect to P_{Z_j} yields $\tau_{Z_j} = \tau$. $\square$

Zero discrepancy and exact balance.

Proof of Proposition 1. The nonnegative random variable $\Delta_j(Z_j; \phi_j)$ has expectation zero exactly when it is zero almost surely. Characteristicness converts zero MMD into equality of the two conditional laws, and overlap makes this equality equivalent to conditional independence. $\square$

Supporting bias decomposition. For this supporting lemma, define $\pi_j(z) \triangleq P(A = 1 \mid Z_j = z)$. For each $a \in \{0, 1\}$ and P_{Z_j} -almost every z , define

$$\delta_{a,j}(z) \triangleq \mathbb{E}_{Q_{1,j,z}}\{\mu_{a,j,z}(V_j, U)\} - \mathbb{E}_{Q_{0,j,z}}\{\mu_{a,j,z}(V_j, U)\}. \quad (84)$$

Lemma 3 (Exact bias decomposition). *Under Assumptions 1–3 and equation (20),*

$$\tau_{Z_j} - \tau = \mathbb{E}[\{1 - \pi_j(Z_j)\}\delta_{1,j}(Z_j) + \pi_j(Z_j)\delta_{0,j}(Z_j)]. \quad (85)$$

Proof. Consistency and Lemma 1 give, for P_{Z_j} -almost every z ,

$$m_{a,j}(z) = \mathbb{E}_{Q_{a,j,z}}\{\mu_{a,j,z}(V_j, U)\}. \quad (86)$$

Conditional expectation over A gives

$$\begin{aligned} \mathbb{E}\{Y(a) \mid Z_j = z\} &= \pi_j(z)\mathbb{E}_{Q_{1,j,z}}\{\mu_{a,j,z}\} \\ &\quad + \{1 - \pi_j(z)\}\mathbb{E}_{Q_{0,j,z}}\{\mu_{a,j,z}\}. \end{aligned} \quad (87)$$

Subtract the two conditional expressions, collect terms, and integrate with respect to P_{Z_j} . $\square$

Approximate cross-view bias bound.

Proof of Theorem 2. Lemma 3, the triangle inequality, stability, and equation (31) give

$$|\tau_{Z_j} - \tau| \leq \mathbb{E}[\{1 - \pi_j(Z_j)\}|\delta_{1,j}(Z_j)| + \pi_j(Z_j)|\delta_{0,j}(Z_j)|] \quad (88)$$

$$\leq \Gamma_j \mathbb{E}\{\Delta_j(Z_j; \phi_j)\}. \quad (89)$$

$\square$

B.2. Proofs for Section 4

Uniform discrepancy deviation.

Proof of Proposition 2. Equation (53) and a multiplicative Chernoff bound imply that every candidate treatment-by-stratum cell contains at least $n_{\text{D}} p_{\min} \eta / 2$ observations except on an event of

probability at most $\delta_D/4$. Conditional on the cell assignments, bounded Hilbert-space mean concentration and a union bound over at most $2M_jN_j$ cells give

$$|\widehat{\Delta}_{j,z}(\phi) - \Delta_j(z; \phi)| \leq 2e_{j,n_D}(\delta_D) \quad (90)$$

simultaneously, with failure probability at most $\delta_D/2$. A multinomial L_1 concentration bound gives

$$\sum_z |\widehat{p}_{j,z}(\phi) - p_{j,z}(\phi)| \leq \sqrt{\frac{2[M_j \log 2 + \log(4N_j/\delta_D)]}{n_D}} \quad (91)$$

simultaneously over candidates, with failure probability at most $\delta_D/4$. On the intersection of the three events, use $0 \leq \Delta_j(z; \phi) \leq 2\rho_j$ and the triangle inequality to obtain equation (56). The three failure probabilities sum to δ_D . $\square$

Representation-learning oracle inequality.

Proof of Theorem 3. On the uniform-deviation event,

$$D_j(\widehat{\phi}_j) \leq \widehat{D}_j(\widehat{\phi}_j) + r_{j,n_D}(\delta_D) \quad (92)$$

$$\leq \inf_{\phi \in \mathcal{F}_j} \widehat{D}_j(\phi) + \varepsilon_{\text{opt},j} + r_{j,n_D}(\delta_D) \quad (93)$$

$$\leq \inf_{\phi \in \mathcal{F}_j} D_j(\phi) + 2r_{j,n_D}(\delta_D) + \varepsilon_{\text{opt},j}. \quad (94)$$

This proves equation (59). Conditional on $\mathcal{I}_D$, the selected map is fixed. Theorem 2 and the common stability budget give $|\tau_{\widehat{Z}_j} - \tau| \leq \overline{\Gamma}_j D_j(\widehat{\phi}_j)$. Substitution proves equation (60). $\square$

Exact signed nuisance remainder.

Proof of Proposition 3. Fix $\widehat{Z}_j = z$. Taking the conditional expectation of the score and using equation (62) gives

$$\begin{aligned} \mathbb{E}\{S_j(O, \epsilon_j) \mid \widehat{Z}_j = z\} &= \widehat{m}_{1,j}(z) - \widehat{m}_{0,j}(z) \\ &\quad + \frac{e_j(z)\{m_{1,j}(z) - \widehat{m}_{1,j}(z)\}}{\widehat{e}_j(z)} \\ &\quad - \frac{\{1 - e_j(z)\}\{m_{0,j}(z) - \widehat{m}_{0,j}(z)\}}{1 - \widehat{e}_j(z)}. \end{aligned} \quad (95)$$

Subtract $m_{1,j}(z) - m_{0,j}(z)$ and collect the two nuisance errors. Averaging the resulting identity over $\widehat{Z}_j$ proves equation (66). $\square$

Conditional finite-sample AIPW error.

Proof of Theorem 4. The boundedness and clipping conditions imply

$$|S_j(O, \epsilon_j)| \leq 2B \left(1 + \frac{1}{\eta}\right). \quad (96)$$

Conditional on the two learning samples, Hoeffding's inequality bounds the difference between the evaluation average and its conditional expectation by the first term in equation (71). Proposition 3, Cauchy–Schwarz, and the propensity clipping bound control the absolute conditional remainder by

$$\frac{q_{e,j}(q_{0,j} + q_{1,j})}{\eta}. \quad (97)$$

The triangle inequality proves equation (72). $\square$

End-to-end sensitivity bound.

Proof of Theorem 5. The triangle inequality gives

$$|\hat{\tau}_j^{\text{AIPW}} - \tau| \leq |\hat{\tau}_j^{\text{AIPW}} - \tau_{\hat{Z}_j}| + |\tau_{\hat{Z}_j} - \tau|. \quad (98)$$

Theorem 4 bounds the first term. Theorem 2 and uniform candidate stability bound the second term by $\bar{\Gamma}_j D_j(\hat{\phi}_j)$. Proposition 2 gives

$$D_j(\hat{\phi}_j) \leq \hat{D}_j(\hat{\phi}_j) + r_{j,n_D}(\delta_D), \quad (99)$$

which proves equation (73). Equation (59) proves the oracle form. A union bound over the discrepancy-learning and evaluation events gives the stated joint probability. $\square$

Consistency.

Proof of Corollary 1. The first three conditions make the evaluation-sample and nuisance-product components of $\varepsilon_{\text{AIPW},j}$ converge to zero. Equation (78) makes the stability-weighted representation term vanish. The failure probability of Theorem 5 also converges to zero, which proves convergence in probability. $\square$